(Please write your Enrollment Number) Enrollment No. 0160IT-2021

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech CSE/IT/CSE-AI >, Semester:<4>
(April -May, 2023)

Subject Code: BCS 204	Subject: Design and Analysis of Algorithms

Time :3 Hours Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 (5*4=20)

(a) Algorithms A and B spend exactly $T_A(n) = 5 \cdot n \cdot \log_{10} n$ and $T_B(n) = 25 \cdot n$ microseconds, respectively, for a problem of size n. Which algorithm is better in the Big-Oh sense? For which problem sizes does it outperform the other?

(b) Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N). Design a greedy algorithm to rearrange the position of the integer in the array to maximize the sum. What is the complexity of the algorithm

(c) Explain the backtracking approach of algorithm design with n-queen problem as an example

(d) Find the first occurrence of the pattern ***xyz*** in the text ***dxbyxyzcdxyz*** using KMP algorithm

UNIT-I

Q2 (a) Work out the computational complexity of the following piece of code: (5+5=10)

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for( int i = n; i > 0; i /= 2 )
  { for( int j = 1; j < n; j *= 2 )
    { for( int k = 0; k < n; k += 2 )
      { ... // constant number of operations } } }
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(b) Solve the recurrence $T(n) = 2T(\sqrt{n}) + \log n$.

Q3 (a) Given a sequence of n numbers such that the difference between the consecutive terms is constant. Design a divide and conquer algorithm for finding the missing term. Assume that the first and last elements are always part of the input sequence and the missing number lies between index 1 to n-1. (6+4=10)

For example, Input: [1, 4, 7, 13, 16], Output: The missing term is 10

(b) Analyse the complexity of the algorithm

UNIT-II

Q4 (a) Explain the Greedy algorithm for generating Huffman code with an example. (4+4+2=10)

(b) Write the pseudo code and analyse the complexity.

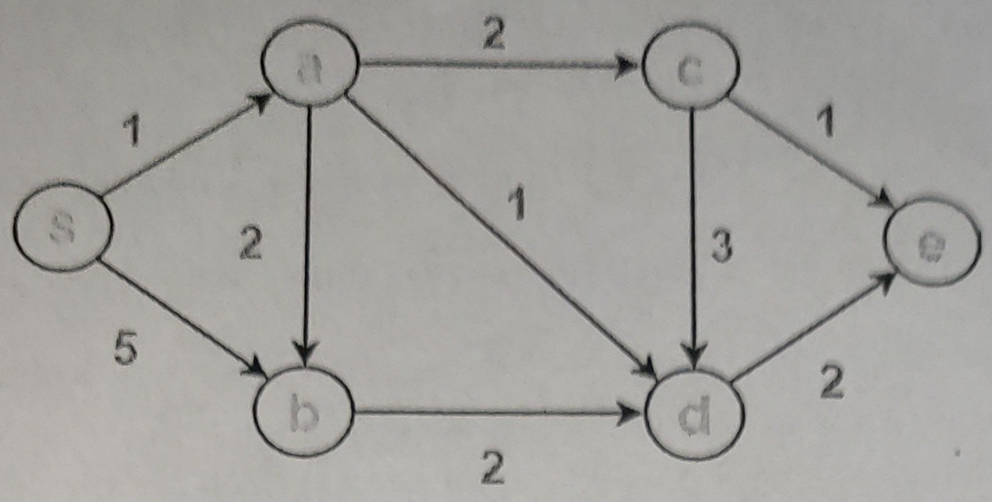
Q5 (a) Explain the Dynamic Programming algorithm for constructing optimal binary search tree (4+4+2=10)

(b) Write the pseudo code and analyse the complexity.

UNIT-III

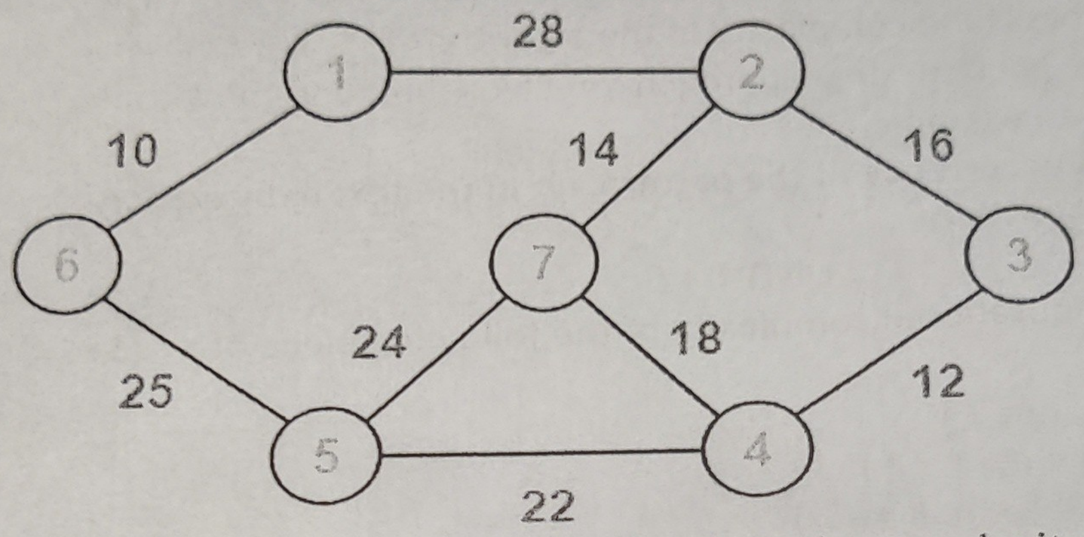
Q6 (a) Find the shortest path from vertex ***a*** to all other vertices in the following graph using Bellman Ford algorithm. (4+4+2=10)

PTO



(b) Write the pseudo code and analyse the complexity.

Q7 (a) Find the minimum spanning tree of the following graph (4+4+2=10)
using Kruskal's algorithm



(b) Write the pseudo code and analyse the complexity.

UNIT-IV

Q8 (a) Explain the Robin Karp algorithm for string matching with an example. (4+4+2=10)

(b) Write the pseudo code and analyse the complexity.

Q9 (a) What are NP-complete problems. (4+6=10)

(b) Prove that set cover problem is NP-complete.